Restraining Orders, Firearm Access, and Lethal Intimate Partner Violence

Asha Bharadwaj* Mary Daniels†

October 22, 2026

Click here for the latest version

Abstract

Firearms are the most common weapon used in intimate partner homicide and sharply raise the probability that intimate partner violence turns lethal. Beginning in the 1990s, many states enacted laws restricting firearm access for individuals subject to a domestic violence restraining order (DVRO). We use the Federal Bureau of Investigation's Supplementary Homicide Reports to measure homicide outcomes and compile data on civil protection order filings from the National Center for State Courts and individual states' annual statistical reports. Using a staggered difference-in-differences design, we find that firearm dispossession laws reduce male-perpetrated firearm homicides of women by approximately 35 percent and intimate partner homicides by roughly 20 percent, with no corresponding increase in homicides involving other weapons. These findings are consistent with the laws operating by disarming the small group of high-risk individuals already subject to restraining orders, rather than by changing aggregate firearm ownership or the demand for court protection. We find no clear evidence that mandatory surrender laws outperform those that leave removal to judicial discretion.

*University of Wisconsin-Madison.
†University of Wisconsin-Madison.

1 Introduction

Firearms substantially increase the lethality risks involved with intimate partner violence (IPV). Researchers have found that an abuser having access to firearms is associated with a 7-fold increase in the risk of intimate partner homicide (Campbell et al., 2003). Moreover, from 1980 to 2020, on average, over fifty percent of intimate partner female homicides in the United States were committed with a firearm (Supplementary Homicide Reports). Recognizing the risks posed by firearms, federal and state lawmakers have passed laws restricting access to firearms for domestic abusers. In particular, beginning in the 1990s, numerous states have passed laws restricting firearm possession and purchase for people subject to a domestic violence restraining order (DVRO) or convicted of a domestic violence misdemeanor.

This paper studies how firearm dispossession laws shape IPV outcomes. We focus on three questions. First, do firearm dispossession laws change the demand for protection orders? Second, do firearm dispossession laws reduce intimate partner homicides? Third, does discretion and enforcement design affect these outcomes? States vary widely in statutory design: some states mandate that firearms be surrendered upon issuance of a DVRO, while others grant judges discretion to decide if firearms should be surrendered. We test whether these differences in policy design translate into different patterns of protection order filings and IPV outcomes.

We use separate data sources for each set of outcomes. For homicide outcomes, we use the Federal Bureau of Investigation's Supplementary Homicide Reports (SHR). For protection order outcomes, we compile state-level records on protection order filings from the National Center for State Courts and individual states' annual statistical reports. We also examine a state-level measure of household firearm ownership from RAND. We link these outcomes to the timing and statutory design of firearm dispossession laws documented by Zeoli et al. (2019). Exploiting variation in the timing of adoption across states, we use a staggered difference-in-differences design that compares outcomes in adopting states with those in states that have not yet adopted a law or never do.

We find that firearm dispossession laws reduce lethal violence. Adoption reduces male-perpetrated firearm homicides of women by approximately 35 percent and intimate partner homicides by roughly 20 percent. These reductions are concentrated in firearm homicides, with no offsetting increase in homicides involving other weapons, indicating that the laws prevent killings rather than shift them to other means. At the same time, we find no detectable effect on state-level household firearm ownership or protection order filings. Taken together, these findings are consistent with a targeted disarmament mechanism: the individuals directly affected by these laws constitute too small a share of the population for their disarmament to register in aggregate firearm-ownership rates, but restricting firearm access among this high-risk group can substantially reduce lethal violence. When we distinguish between mandatory- and discretionary-surrender laws, the estimates are negative for both groups but too imprecise to establish that one design is more effective. This comparison is particularly underpowered because relatively few states leave surrender to judicial discretion.

1.1 Literature Review

This paper makes several contributions to the literature on the effectiveness of firearm dispossession laws for individuals subject to domestic violence restraining orders. First, we extend the work of Vigdor and Mercy (2006) and Zeoli and Webster (2010) by estimating the effect of firearm dispossession laws enacted in the 2000s and addressing some of their methodological and data limitations. Zeoli and Webster (2010) provide descriptive evidence that firearm dispossession laws for people subject to DVROs enacted between 1979 and 2003 are associated with a decrease in intimate partner homicide, and Vigdor and Mercy (2006) estimate a two-way fixed effects model to provide evidence that there is a causal relationship between these variables. Given the potentially important impact these laws can have on homicide rates, it is valuable to have multiple causal studies examining their effectiveness.

Moreover, we improve upon these papers in numerous ways. First, we address concerns about how heterogeneous treatment effects can contaminate difference-in-differences esti-

mates (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021) by using the robust estimators created by Callaway and Sant'Anna (2021) and Wing et al. (2024) to estimate the effect of firearm dispossession laws on intimate partner and female homicide. Second, there is a long-standing concern that the SHR underestimates the number of IPH incidents that occur because it fails to identify former dating partners, which Zeoli and Webster (2010) and Vigdor and Mercy (2006) do not address. We account for this issue by estimating the effect of firearm dispossession laws on male-perpetrated female homicides, which we argue serve as a good proxy for IPH. Third, we address the potential confounding impact of the Brady Handgun Violence Prevention Act of 1993 by focusing on estimating the treatment effect of state-level firearm dispossession laws that were passed from 2000 to 2016. This differs from the aforementioned papers, which primarily focus on states that enacted firearm dispossession laws in the 1990s.

Next, we introduce protection order filings as an outcome for evaluating firearm dispossession laws. Existing evaluations focus on intimate partner homicide and related violence; to our knowledge, we are the first to study the laws' effect on the demand for protection orders. Lynch et al. (2022) find that among women in a domestic violence shelter, those who had experienced firearm abuse were more likely to seek a protection order than those who had not. If firearm abuse raises the demand for orders, laws that restrict abusers' access to firearms may change it. We assemble a state-year panel of protection order filings and estimate whether adopting a dispossession law affects the number of orders sought.

More broadly, we contribute to the literature examining how interventions by the legal system impact intimate partner violence. Holt et al. (2002) and Holt et al. (2003) find that domestic violence victims who obtain a permanent civil protection order are less likely to report subsequent abuse 5–12 months after getting the order compared to women who did not obtain a civil protection order. Aizer and Bó (2009) find that no-drop prosecution policies increase reporting of domestic violence and reduce intimate partner homicides of men. We contribute to this literature by examining the effectiveness of firearm dispossession laws on

intimate partner homicide.

2 Motivation

This section documents three features of intimate partner homicide that motivate the policy we study and our empirical strategy:

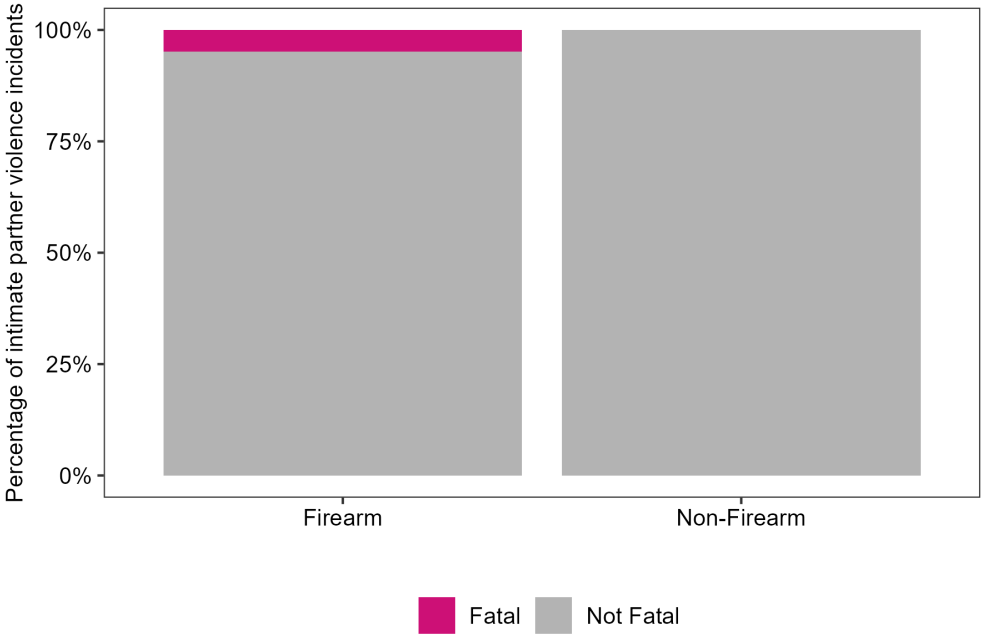
1. Firearms raise the probability that intimate partner violence is fatal;
2. Firearms are the weapon used in most intimate partner homicides; and
3. The victims of firearm intimate partner homicides are almost always women.

We define an intimate partner homicide as a homicide committed by a current or former spouse or dating partner.

Firearms raise the probability that intimate partner violence is fatal. Most intimate partner violence is not lethal, but whether it becomes lethal depends heavily on the weapon involved. Figure 1 reports the case-fatality rate, which is the share of incidents in which the victim was killed, by weapon. Among incidents involving a firearm, 4.8 percent ended in the victim's death; among incidents involving any other weapon, fewer than 1 percent did. This difference in case-fatality rates motivates the policy we study, which restricts firearm access for individuals subject to a domestic violence restraining order.

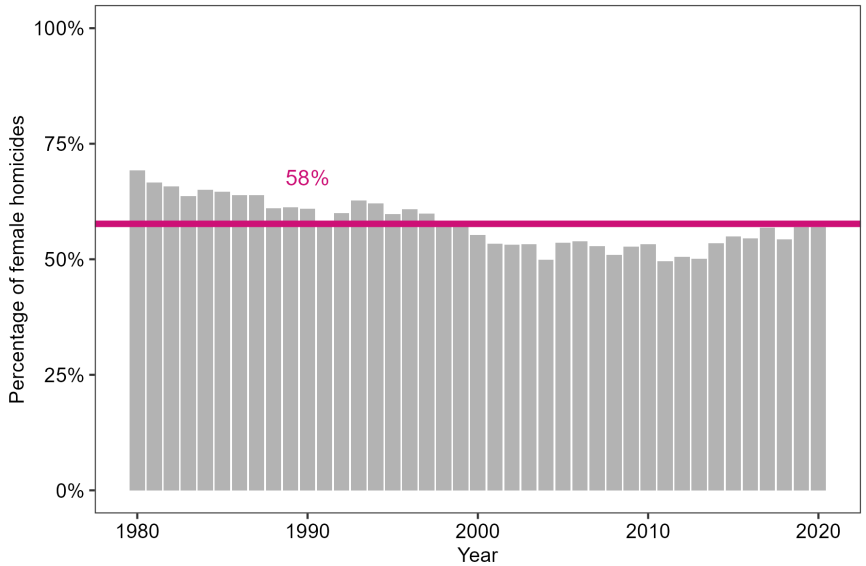
Most intimate partner homicides are committed with a firearm. More than 75 percent of all homicides in the United States between 2018 and 2022 were committed with a firearm (Harrell et al., 2024). Intimate partner homicide is no exception: in every year from 1980 to 2020, firearms were used in at least half of reported intimate partner homicides, and in 58 percent on average (Figure 2).

Figure 1: Case-Fatality Rate of Intimate Partner Violence, by Weapon



Source: Own calculations based on the FBI's National Incident-Based Reporting System, 1991-2016.
Notes: The case-fatality rate is the share of intimate partner violence incidents resulting in the victim's death, computed separately by the weapon involved.

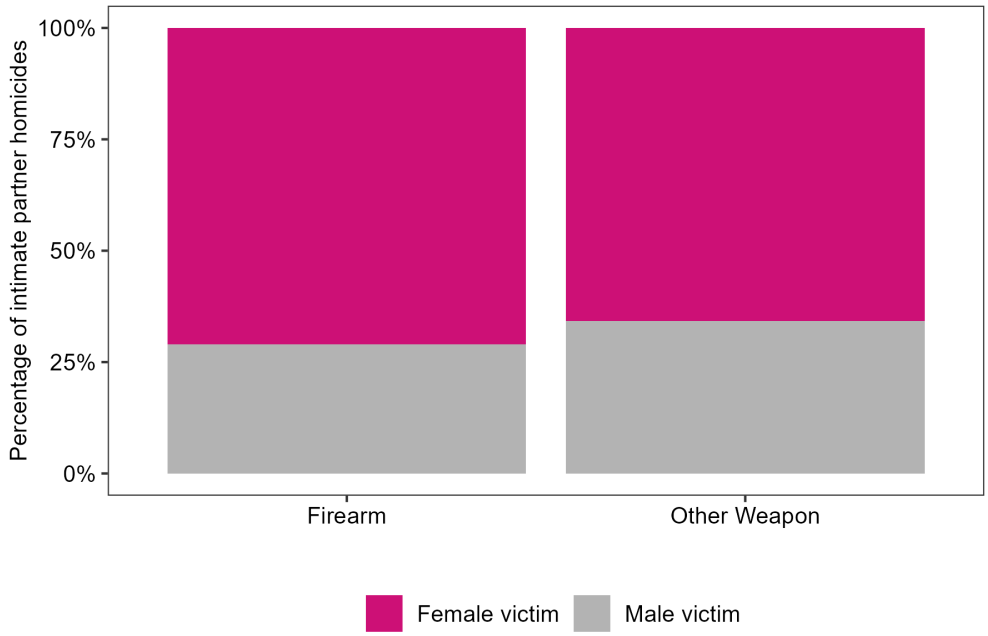
Figure 2: Firearm Usage in Intimate Partner Homicides, 1980–2020



Source: Own calculations based on data from the Uniform Crime Reports: Supplementary Homicide Reports, 1980–2020.
Notes: Intimate partners include current dating partners and current and former spouses. The horizontal line reports the mean annual percentage of intimate partner homicides that involved the use of a firearm.

The majority of victims are women. Figure 3 shows that the large majority of intimate partner homicides have a female victim, and that this holds whether or not a firearm is used. Accordingly, in our analysis, we focus on male-perpetrated female homicide as one of our main outcomes.

Figure 3: Intimate Partner Homicide by Victim Sex, 1980–2020



Source: Own calculations based on the FBI Supplementary Homicide Reports, 1980–2020.
Notes: Intimate partners include current dating partners and current and former spouses.

Together, these facts point to firearm access among abusers as a target for policy. The next section describes the federal and state laws that restrict it.

3 Institutional Background

Firearms both raise the lethality of intimate partner violence and are the weapon used in most intimate partner homicides. In response, Congress and many state governments have enacted laws restricting firearm access for individuals who pose a heightened risk of domestic homicide, including those with a history of domestic abuse. In this paper, we focus on laws restricting access to firearms for individuals subject to a domestic violence restraining

order (DVRO). A DVRO is a civil court order intended to protect an individual from abuse and/or harassment from someone with whom they have a domestic relationship. Domestic relationships generally include intimate partners, family members and people who share a child in common.

There are two types of DVROs: ex parte/temporary orders and full orders. Ex parte orders are issued for a limited amount of time (usually no longer than a week) and do not require the respondent to be notified or given a chance to respond to the allegations before they can be issued. Full orders are issued for a longer period of time (usually a year) and can only be issued after a hearing where both parties are given the opportunity to participate (see Appendix Table 9 for more information on DVRO terminology). Nationally, civil protection orders were filed at a rate of roughly 380 per 100,000 residents per year over 2004–2010.[1]

The 1996 Lautenberg Amendment to the Federal Gun Control Act made it illegal under federal law for anyone convicted of a domestic violence misdemeanor under federal, state or tribal law to possess, ship, or transport firearms or ammunition by way of interstate or foreign commerce.[2] The 1994 Violence Against Women Act also made it illegal under federal law for those subject to a final domestic violence restraining order to possess, ship, or transport firearms or ammunition.[3]

The federal prohibition, while broad in scope, does little on its own to disarm the people it covers. There is no affirmative requirement that respondents who already possess a firearm turn in their weapons to law enforcement or a federal firearms licensee, and no process for the courts or law enforcement to recover them. Beginning in the 1990s, many states passed their own laws governing firearm possession and/or purchasing for people subject to a DVRO or convicted of a domestic violence misdemeanor. Some of these states require respondents to surrender firearms they already possess, while other states leave dispossession to the court's discretion (Zeoli et al., 2019). Whether an abuser subject to a restraining order is actually

[1] Own calculations from the NCSC Court Statistics Project, covering 36–46 reporting states annually.
[2] 18 U.S.C. § 922(g)(9)
[3] 18 U.S.C. § 922(g)(8)

disarmed therefore depends on state law rather than the federal prohibition, which is why our analysis focuses on the adoption of state dispossession laws.

State laws also reach further than the federal prohibition in two respects. First, whereas federal law reaches only orders issued after notice and a hearing, many states extend firearm restrictions to ex parte orders as well.[4] Second, the federal prohibition applies only to current or former spouses, cohabitants, and those who share a child in common,[5] while many states also allow dating partners who never cohabited to seek an order and the associated restrictions.

There is substantial heterogeneity in the way states design firearm dispossession laws. As of February 2016, 29 states and the District of Columbia had laws on the relinquishment or removal of firearms for individuals subject to a DVRO (Zeoli et al., 2019). Nearly all (28 states and the District of Columbia) instruct courts on how to execute the firearm dispossession law for people subject to full DVROs (Table 1).[6] Of those with instructions for full DVROs, 16 (57%) permit or require dispossession for people subject to ex parte (temporary) DVROs. About 40% of states give judges discretion to determine if a DVRO respondent cannot possess a firearm while the remaining states *require* judges to order dispossession.[7]

4 Data

This section describes the data used for this paper. We primarily rely on two data sets: (1) records of civil protection orders and (2) the Federal Bureau of Investigation's Supplementary Homicide Reports (SHR) for 1990 to 2016.

[4]Sixteen states permit or require dispossession for respondents subject to an ex parte order (Zeoli et al. 2019).

[5]18 U.S.C. § 921(a)(32)

[6]As of September 2025, 42 states have a law prohibiting people subject to DVROs from owning, possessing, or purchasing firearms (Corporation, 2025).

[7]Washington state requires dispossession in certain instances and gives judges discretion in others. Judges *shall* order dispossession if there is "clear and convincing evidence use, display, threat with gun, or dangerous weapon in a felony; previous gun restrictions; or where the order meets the federal prohibition," but *may* order if "there is a preponderance of evidence of use, display, threat with gun, or dangerous weapon in a felony or previous gun restrictions" (Zeoli et al., 2019). For the purposes of this paper, Washington is categorized as requiring dispossession.

Table 1: Statistics on Firearm Dispossession Laws

	Type of DVRO	
	Ex parte	Full
Number of states providing instructions to the courts	16	29
Judges have discretion to determine if a DVRO respondent is prohibited from possessing firearms	50%	37.9%
Law instructs courts on how to facilitate dispossession for newly prohibited respondents	18.8%	17.2%
Law includes a provision specifying the court's authority to order law enforcement to search and seize firearms	31.2%	20.7%
Dispossession includes firearm purchase permits and/or carry permits/licenses	43.8%	27.6%
Order issued by the court must include information about firearm dispossession	25%	24.1%

Source: Own calculations based on data from Zeoli et al. (2019).

Notes: This table presents statistics on the design of firearm dispossession laws. It summarizes laws in 28 states and the District of Columbia that prohibit individuals subject to either ex parte or full domestic violence restraining orders (DVRO) from possessing firearms.

Supplementary Homicide Reports (SHR)

The Federal Bureau of Investigation (FBI) collects data on crime from local law enforcement agencies through the Uniform Crime Reporting Program. A product of this program is the SHR. This is a collection of crime incident reports from law enforcement agencies that documents the details of incidents identified as murder, non-negligent manslaughter, and negligent manslaughter. Each incident report documents the number of victims and offenders (up to eleven of each), the demographics of the victim(s) and offender(s), the relationship between the first victim and offender(s), the weapon(s) used, and the circumstances surrounding the homicide (e.g., victim shot during hunting accident).

There are three noteworthy limitations to the SHR. First, reporting to the SHR is not mandatory in all states, resulting in underestimates of national crime rates. For example, our calculations found only 3,042 out of 22,645 law enforcement agencies in the country reported data to the SHR in 2016. While the reporting agencies span 49 states and the District of Columbia, they only represent 59.0% of the U.S. population. Second, SHR does not have a category for ex-boyfriends/ex-girlfriends in the variable that identifies the relationship between the victim and offender. One multi-city study found that around 20 percent of all intimate partner homicides of women are perpetrated by an ex-boyfriend (Campbell et al., 2003), suggesting IPH estimates constructed from the SHR may face substantial measurement error. Third, when there are multiple victims, SHR only reports the relationship between the first victim and the offenders (Kaplan, 2024). Therefore, if a law enforcement agency does not list the offender's intimate partner as the first victim, there may be an under-count of the number of IPH incidents. However, Kaplan (2024) estimates that 96% of SHR incidents from 1976 to 2022 involved one victim, so this problem is not likely to introduce substantial measurement error.

Civil Protection Orders

We assembled a state-by-year panel of civil protection order filings. The state panel draws on two sources that report protection orders differently. The Court Statistics Project, compiled by the National Center for State Courts, reports a single count of all civil protection orders for each state and covers most states from 2003 through 2010. Separately, individual state courts publish annual statistical reports that tabulate the specific protection order category defined by state law, such as orders for protection in Minnesota or relief-from-abuse orders in Vermont; in several states these reports run from the early 2000s through the early 2020s. We collect the state reports for the states that publish them over a period long enough to bracket a change in law.

These two sources yield two measures. From the Court Statistics Project, we construct a broad measure that counts all civil protection orders. It is available for 48 states, almost entirely over 2003 through 2010. From the state reports we construct a domestic violence measure that counts petitions under the state's domestic violence statute. It is available for 10 states, in most cases over a longer span. Both measures count petitions filed.

The treated states adopt their laws between 2005 and 2016. Identifying the effect of a law requires observing filings both before and after its adoption, and the Court Statistics Project, which ends in 2010, does not cover the post-adoption years of the states that adopt later. For those states we rely on the domestic violence measure from their own court reports, which extends beyond 2010. The Court Statistics Project supplies the non-adopting states that form the comparison group. We assign each state a single source for the full period, so that within-state variation in filings reflects protection order activity rather than a change in measurement.

Covariates

We obtained data on covariates from various sources. The FBI's UCR Police Employee Data were used to obtain the number of police employees per 10,000 people and the share of

officers who are female for each law enforcement agency (LEA). Following Koppa (2024), we use the number of police employees per 10,000 people to control for differences in each city's capacity to combat IPV. Miller and Segal (2019) find that as the share of female officers in a county increases, reporting of domestic violence increases, while IPH decreases. Thus, LEAs who experience an increase in the share of officers who are female may see improvements in IPH rates that are independent of the dispossession laws. Accordingly, the share of officers who are female is included as a covariate.

We also account for sociodemographic factors that are correlated with IPV. Data show that women make up the overwhelming majority of domestic violence victims (Criminal Justice Information Services Division, 2026), non-Hispanic black women experience higher rates and risks of IPV than non-Hispanic white women (Billings and Daniels, 2026; Capaldi et al., 2012), unemployment is a risk factor for IPH (Campbell et al., 2003, 2007), and IPV is correlated with alcohol use (Billings and Daniels, 2026; Campbell et al., 2007; Sharps et al., 2003). Thus, we obtained county-level data on the share of a county's population that is female and non-Hispanic white from the U.S. Census Bureau's Intercensal Estimates, the county-level unemployment rate from the U.S. Bureau of Labor Statistics, and state-level data on alcohol consumption from the National Institute on Alcohol Abuse and Alcoholism (Kaplan, 2019).

4.1 Sample selection

We restrict the treated group to states that adopted a firearm dispossession law after 2000. Our reason is institutional. Before the Brady Handgun Violence Prevention Act of 1993, no national system existed to screen firearm purchasers, so a prohibited individual could readily acquire a gun from a licensed dealer. The Brady Act sought to close this gap by mandating background checks, but its enforcement mechanism took years to mature. The Act's interim provisions (in effect 1994–1998) relied on a waiting period during which local law enforcement were required to conduct the checks. In Printz v. United States (1997), the

Supreme Court struck down that mandate as an unconstitutional commandeering of state officials, shifting responsibility to the federal government. In response, the FBI built the National Instant Criminal Background Check System (NICS), which launched on November 30, 1998 and extended instant, point-of-sale verification to all firearm purchases from licensed dealers nationwide (Campbell, 2022). Until NICS was operational, dispossessing a prohibited individual of a firearm would have done little to restrict access, since no system reliably prevented that individual from simply buying another. By focusing on states that adopted dispossession laws after 2000, we study these laws during a period when the infrastructure needed to enforce them was in place.

Table 2 reports summary statistics for our analytical sample. Panel A presents the outcome variables, all expressed as rates per 100,000 population. On average, a county-year experiences 0.82 male-perpetrated female homicides by any weapon, of which 0.46 involve firearms; intimate partner homicides average 0.66 overall and 0.39 by firearm. All four outcomes are right-skewed, with means exceeding medians and minima of zero, reflecting the relative rarity of these events at the county-year level. Panel B summarizes demographic characteristics. The sample is balanced by sex (51% female), predominantly non-Hispanic White (mean of 65%, though ranging widely from 3% to 98%), and reports a mean per-capita ethanol consumption of 2.20 gallons and a mean unemployment rate of 5.87%. Panel C describes policing controls: police employees average 27.1 per 10,000 people, and women constitute 11% of officers on average. Panel D characterizes the contemporaneous policy environment. Domestic-violence arrest regimes vary across the sample, with 46% of state-years under mandatory-arrest laws, 7% under preferred-arrest, and 39% under discretionary-arrest rules; 12% have no-drop prosecution policies. Firearm policies beyond dispossession are also present, with 12% of state-years subject to universal background checks and 11% requiring a permit to purchase a firearm.

Table 2: Summary statistics

	Mean	Median	SD	Min	Max
Panel A: Outcomes					
Male-Perp. female homicides, all weapons	0.82	0.58	0.99	0.00	11.26
Intimate partner homicides, all weapons	0.66	0.39	0.90	0.00	8.44
Male-Perp. female homicides, firearms	0.46	0.15	0.70	0.00	6.53
Intimate partner homicides, firearms	0.39	0.00	0.67	0.00	6.87
Panel B: Demographic controls					
Share female	0.51	0.51	0.01	0.46	0.55
Share non-Hispanic White	0.65	0.68	0.18	0.03	0.98
Ethanol consumption per capita	2.20	2.21	0.28	1.19	4.78
Unemployment rate	5.87	5.40	2.39	1.20	22.50
Panel C: Policing controls					
Police employees per 10,000 people	27.14	25.89	9.78	0.00	110.12
Share of female officers	0.11	0.11	0.06	0.00	0.93
Panel D: Contemporaneous policies					
Mandatory arrest	0.46	0.00	0.50	0.00	1.00
Preferred arrest	0.07	0.00	0.25	0.00	1.00
Discretionary arrest	0.39	0.00	0.49	0.00	1.00
No drop prosecution	0.12	0.00	0.33	0.00	1.00
Universal background checks	0.12	0.00	0.32	0.00	1.00
Permit to purchase	0.11	0.00	0.31	0.00	1.00

Protection Orders. Protection order filings are observed at the state-year level for 23 states over 2003–2018, of which six adopt a dispossession law during the period. A state-year records 13,731 filings on average, with a median of 10,876 and a range from 48 to 54,446 across 189 state-years. The wide range reflects differences in state population and in the breadth of each state's protection order category. Filings enter the analysis in logs.

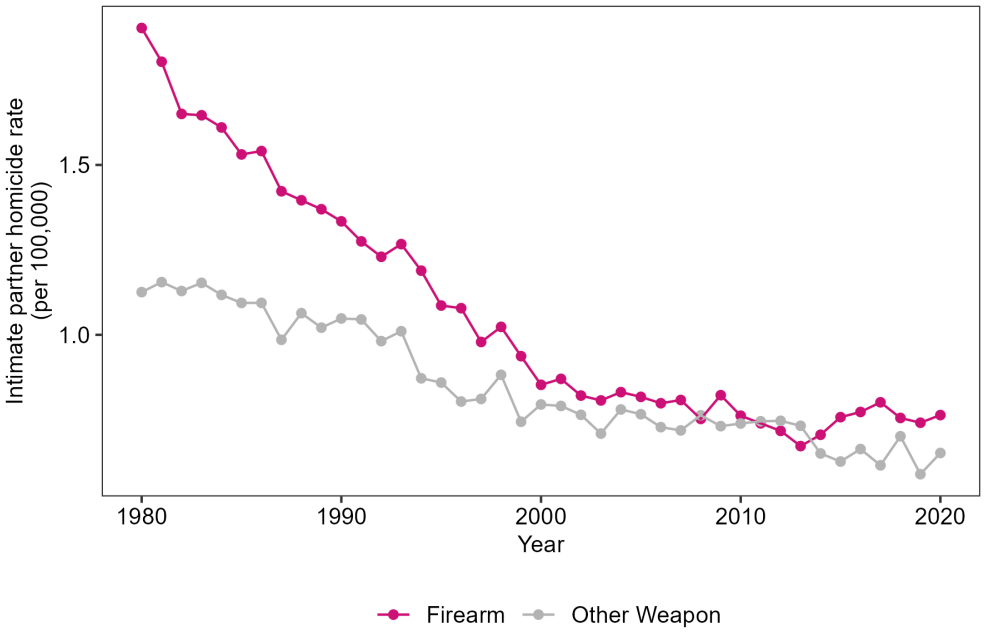
5 Empirical Strategy

A natural first approach would be to regress intimate partner homicide on an indicator for whether a state has a firearm dispossession law in effect, pooling across states and years. This would not recover the causal effect of the law, because whether and when a state

adopts one is not random. States choose to enact these laws, and that choice is plausibly correlated with determinants of intimate partner homicide that we do not observe. States with stronger support for gun regulation, lower firearm ownership, or distinctive attitudes toward domestic violence may be both more likely to adopt a dispossession law and different in their underlying rates of lethal violence. A cross-sectional comparison of states with and without the law would confound the law's effect with these pre-existing differences, and the direction of the bias is not obvious: if adopting states already had lower homicide for other reasons, the comparison would overstate the law's benefit; if states adopt in response to a domestic violence problem, it would understate it.

Comparing intimate partner homicide before and after adoption within adopting states does not solve the problem either. Intimate partner homicide, and firearm homicide in particular, declined across the country from 1980 to 2016 (Figure 4). A before-and-after comparison would attribute this common national trend to the dispossession laws.

Figure 4: Intimate Partner Homicide Rate by Weapon, 1980–2020



Source: Own calculations based on the FBI Supplementary Homicide Reports, 1980–2020.
Notes: Rates are intimate partner homicides per 100,000 population, plotted separately for firearm and non-firearm homicides.

Our difference-in-differences design addresses both problems. County fixed effects absorb time-invariant differences across places, so the law's effect is identified from changes within a county rather than from comparisons of levels across counties. Year fixed effects absorb shocks common to all counties in a given year, including the national trend in homicide. Identification then rests on the weaker assumption that, absent the law, homicide in adopting and comparison areas would have followed parallel paths—the assumption we examine in Section 5.1. We estimate the following equation:

$$Y_{ct} = \beta_1 \, Firearm_{st} + \mu X_{cst} + \gamma_c + \tau_t + \varepsilon_{ct} \tag{1}$$

where Y_{ct} is the outcome of interest for county c in year t, $Firearm_{st}$ is an indicator equal to 1 if state s had a firearm dispossession law in year t, X_{cst} is a vector of time-varying, county-level covariates, γ_c is county fixed effects, and τ_t is year fixed effects. The coefficient of interest is β_1, which reports the average treatment effect on the treated. Standard errors are clustered by state, the level of treatment.

We examine outcomes related to intimate partner homicide. Our primary outcomes are intimate partner homicides and firearm intimate partner homicides per 100,000 people. As discussed earlier, SHR's failure to identify homicides committed by a former dating partner may introduce substantial measurement error into our estimates of IPH rates. Thus, we also include male-perpetrated female homicides and firearm male-perpetrated female homicides per 100,000 as outcomes. Since the majority of domestic violence victims are female and the majority of offenders are male, this should permit us to capture most IPH committed by a former dating partner (Criminal Justice Information Services Division, 2026).

We control for a variety of demographic characteristics, policing characteristics, and other policy changes that could affect IPH and violence against women. Since women make up the overwhelming majority of victims of domestic violence (Criminal Justice Information Services Division, 2026), we control for the percentage of the population that is female. We also control for the percentage of the population that is non-Hispanic white and ethanol consumption per

capita, since race and alcohol use are correlated with domestic violence (Billings and Daniels, 2026; Criminal Justice Information Services Division, 2026). To account for the potential impact of policing on domestic violence, we control for the number of police employees per 10,000 and the share of officers that are female.

Additionally, we control for the presence of other firearm and intimate partner violence-related laws and policies. Specifically, we include indicators equal to 1 if in year t, state s had (1) a law requiring a permit to purchase firearms, (2) a law requiring universal background checks for gun purchase, (3) a mandatory arrest law for domestic violence incidents, (4) a preferred arrest law for domestic violence incidents, and (5) a discretionary arrest law for domestic violence incidents. Using data from Aizer and Bó (2009), we also include an indicator equal to one if a city within county c had a no-drop prosecution policy in year t. Findings from Aizer and Bó (2009) and Chin and Cunningham (2019) indicate that arrest laws and no-drop prosecution policies have statistically significant effects on IPH.

Estimator. A recent literature shows that the two-way fixed effects (TWFE) estimator can fail to recover a sensible average treatment effect under staggered treatment timing. When treatment effects vary across adoption cohorts or over time, TWFE returns a weighted average of effects in which some weights can be negative, in part because it uses already-treated units as controls for later-treated ones (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021). We therefore recover β_1 with the doubly robust estimator of Callaway and Sant'Anna (2021), which compares each treated cohort only to not-yet-treated counties and aggregates the resulting cohort-by-period effects with non-negative weights (see Appendix A.4). As a robustness check, we also report estimates from the stacked difference-in-differences estimator of Wing et al. (2024) in Appendix A.2.

The estimators treat the covariates differently. In the two-way fixed-effects model the covariates enter contemporaneously, whereas the Callaway and Sant'Anna (2021) estimator conditions on their pre-treatment values through the doubly-robust moment condition. Conditioning on base-period rather than contemporaneous values avoids controlling for char-

acteristics that treatment may itself affect, which would otherwise bias the estimate.

5.1 Identifying Assumptions

This section discusses the assumptions we are making to identify the treatment effect of firearm dispossession laws. Our staggered DID design exploits variation in *when* these laws were enacted to identify their treatment effect. We assume that while states choose whether or not to enact these laws, the exact year a law is enacted is as good as random. From 1990 to 2016, 17 states enacted a firearm dispossession law. As shown in Table 3, most of the treatment states enacted these laws in quick succession in the 1990s and early 2000s, a period of widespread gun control actions in the United States (ThoughtCo., 2025). Thus, it is reasonable to believe that while states did consciously decide to enact these laws to match the national trends towards more gun control, they did not pre-select the exact year they would enact a dispossession law.

Table 3: Enactment Dates for Laws Prohibiting Firearm Possession for Individuals Subject to DVROs

Year firearm dispossession law was enacted	Number of states
1991	1
1993	2
1994	3
1996	3
2002	2
2005	1
2007	1
2009	1
2010	1
2014	1
2016	1
Never enacted	18

Conditional Parallel Trends. The key identifying assumption for a DID model is that absent treatment, the treatment and control groups would have exhibited parallel trends in the outcome in the post-treatment period. While this assumption is not directly testable, we use an event study model to determine if it is plausible. The event study model is

$$Y_{ct} = \sum_{\substack{l=-K \\ l\neq -1}}^{l=L} \beta_l \, D_{c,t}^{l} + \mu X_{cst} + \gamma_c + \tau_t + \varepsilon_{ct} \tag{2}$$

where $D_{c,t}^{l}$ is an indicator equal to 1 if in year t, county c is l years from when their state enacted a firearm dispossession law, X_{cst} is a vector of time-varying, county-level covariates, γ_c is county fixed effects, and τ_t is year fixed effects. These event study indicators report the effect of having a firearm dispossession law in l years after enactment, conditional on the covariates X_{cst}. If the pre-treatment coefficients (β_l for $l \in \{-K, \ldots, -2\}$) are constant, relatively close to zero, and statistically insignificant, we can conclude that the treatment and control groups exhibited parallel trends in the outcomes in the pre-treatment period. Therefore, it is reasonable to assume that absent treatment, they would have continued to exhibit parallel trends in the post-treatment period. We estimate these dynamic effects with the Callaway and Sant'Anna estimator, whose event-study aggregation is the analogue of β_l (Appendix A.4).

No Anticipation Effects. The second identifying assumption is that there were no treatment effects in the treated counties prior to the enactment of firearm dispossession laws. This would occur if crime victims, offenders, and/or criminal justice professionals changed their behavior in anticipation of the enactment of firearm dispossession laws. The plausibility of this assumption can also be determined by examining the event coefficients. Since there should not be a treatment effect prior to the enactment of a law, the coefficients should be near zero and statistically insignificant prior to treatment and exhibit no clear change until after treatment.

Balance Tests. Lastly, we test for the presence of confounding variables. In a difference-in-differences design, identification stems from the parallel trends assumption detailed earlier. The credibility of this assumption is strengthened if researchers can show that there are no other confounding variables that provide alternative explanations for the observed treatment effect. We do this by conducting balance tests. First, we test if the pre-treatment means for the treated and never-treated states were statistically different from one another. If they are not, this makes it more believable that they would have trended similarly in the absence of treatment. Second, we conduct the left-hand side balance test detailed by Pei et al. (2019), which estimates Equation 1 using each covariate as the dependent variable and excluding the vector of time-varying covariates X_{cst}. If the timing of treatment (enacting a dispossession law) is uncorrelated with changes in the covariates, then we can assume any observed treatment effect is truly due to the enactment of a firearm dispossession law.

Table 4 reports the results of the balance tests. Columns (1)-(3) report the pre-treatment means for the treatment and control groups, while column (4) reports the average treatment effect of firearm dispossession laws on the covariates. Overall, the treatment and control groups exhibited good pre-treatment balance. The only statistically significant differences were that the control group had a higher Hispanic population share, share of the population under age 35, and unemployment rate. More importantly, the enactment of firearm dispossession laws was correlated with increases in the percentage of the population that is under age 35 and the unemployment rate. Youth and unemployment are both positively correlated with IPV, suggesting that these sociodemographic changes would *increase* homicides, not decrease them. Thus, we do not believe these changes present a threat to our identification strategy.

Table 4: Balance Tests

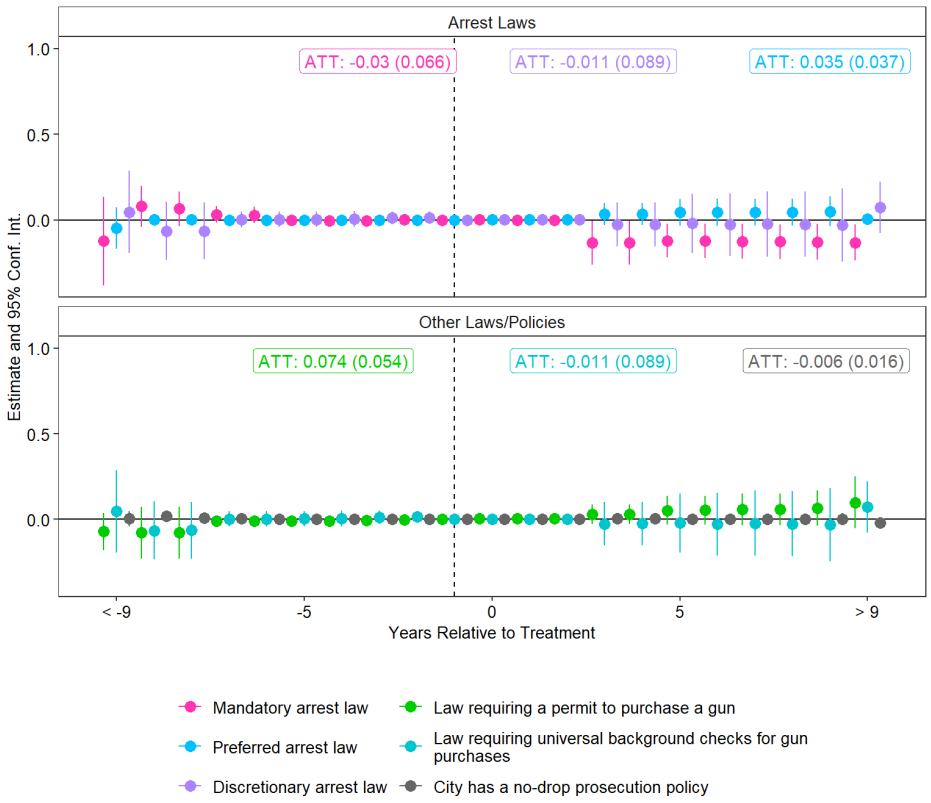
	Pre-treatment means, 1990			Left-hand side test
	Treatment	Control	p-value of difference between (1) and (2)	ATT
	(1)	(2)	(3)	(4)
Pct Non-Hispanic white	0.74	0.71	0.261	-0.009
				(0.011)
Pct Non-Hispanic black	0.21	0.19	0.646	0.003
				(0.007)
Pct Hispanic	0.03	0.07	0.003	0.006
				(0.009)
Pct under age 35	0.55	0.56	0.019	0.008**
				(0.003)
Pct female	0.52	0.52	0.537	0.000
				(0.001)
Unemployment rate	4.86	6.08	0.000	0.963***
				(0.300)
Police employees per 10,000	26.63	23.57	0.139	-2.302
				(2.462)
Female share of officers	0.09	0.09	0.719	-0.003
				(0.009)
Ethanol consumption (gallons per capita)	2.29	2.36	0.400	0.060
				(0.054)
Robberies per 100,000	193.62	194.24	0.982	-7.662
				(14.463)
Observations	47	150		5,713
County Fixed Effects				Yes
Year Fixed Effects				Yes

***p<0.01, **p<0.05, *p<0.1

Notes: This table reports the results of two balance tests conducted on the county-level homicide data set constructed for this paper. Columns (1)-(2) report the pre-treatment means of the covariates for the treatment and control groups. Column (3) reports the p-value from two-sample t-tests with the null hypothesis that the means of the treatment and control groups are equal. Column (4) reports estimates of the average treatment effect of firearm dispossession laws on the covariates. Estimates were obtained from a two-way fixed effects model regressing the covariate of interest on an indicator equal to 1 if county c had a firearm dispossession law in year t, county fixed effects, and year fixed effects. The standard errors were clustered by state and are reported in parentheses.

Concurrent policy changes

Figure 5: The Relationship Between Firearm Dispossession Laws and Other Laws/Policies



Notes: This figure plots event study estimates and 95 percent confidence intervals from regressions estimating the effect of firearm dispossession laws on other state-level arrest laws for domestic violence incidents, state-level gun laws, and city-level no-drop prosecution policies. For each state-level law, an indicator equal to 1 if the law was effective in county c in year t was created. For the no-drop prosecution policies, an indicator was created that was equal to 1 if there was a city in county c with a no-drop prosecution policy in year t. Then, estimates were obtained from a two-way fixed effects model regressing the indicator for the law/policy of interest on event time indicators equal to 1 if county c is l years from when their state enacted a firearm dispossession law, county fixed effects, and year fixed effects. Estimates of the average treatment effect on the treated (ATT) are reported at the top of figures. The standard errors are reported in parentheses and were clustered by state.

As mentioned earlier, firearm dispossession laws were enacted during a period when many firearm and domestic violence-related laws were being enacted across the United States. The enactment of these other laws could pose a threat to identification if they were adopted in the same years as dispossession laws, or if their timing were otherwise correlated with it. We

tested this by conducting the left-hand side balance test. The results show that within our data set, firearm dispossession laws were uncorrelated with the presence of other firearm- and domestic-violence related laws (Figure 5).

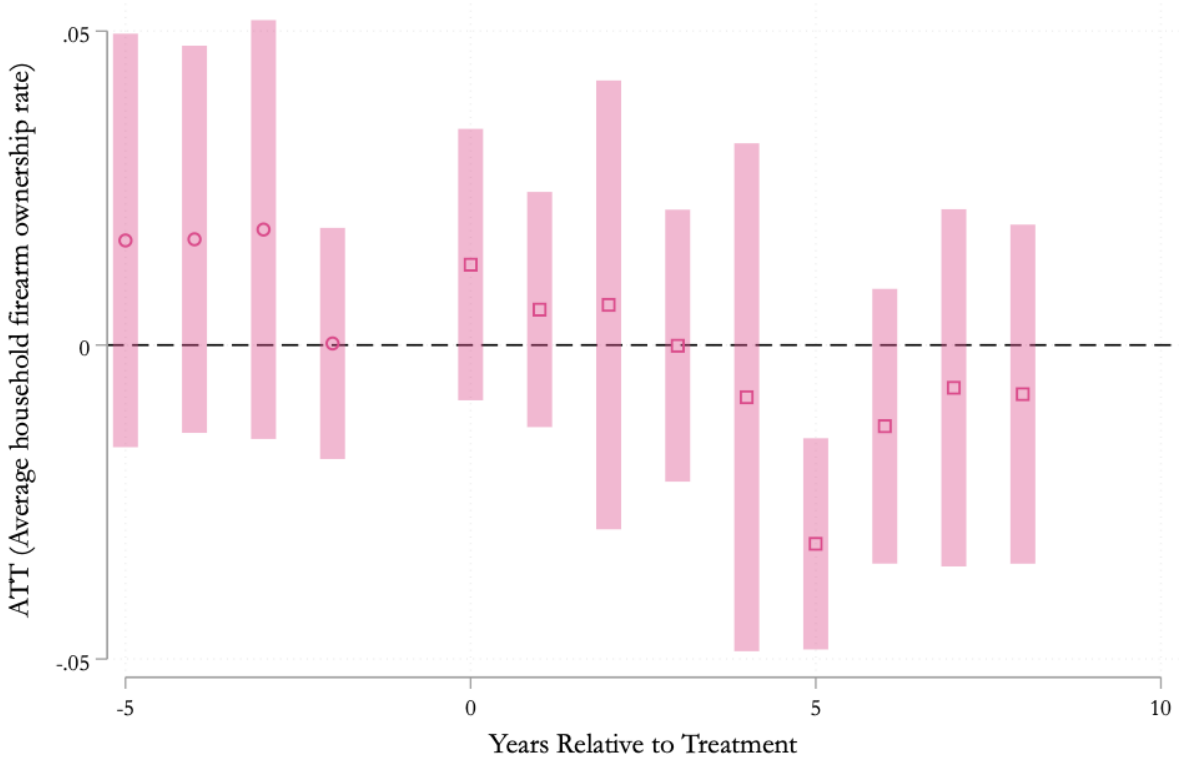
6 Results

6.1 Effects on household firearm ownership

We begin with the effect of firearm dispossession laws on household firearm ownership. This outcome is first-order for our analysis because it is the margin on which the policy operates most directly: a dispossession provision tied to a restraining order requires a prohibited individual to surrender or forgo firearms, so its immediate, intended consequence is a reduction in gun access. Examining ownership directly thus tells us whether the policy does what it is designed to do before examining its effect on lethal violence.

A limitation of our data is that firearm ownership is only observed at the state level. Since the dispossession laws only target individuals subject to a restraining order, a statewide aggregate may be too coarse to detect disarmament concentrated within a small, targeted group, so a null result on this measure does not rule out that the law removes firearms from the individuals it covers.

Figure 6: Effect of Firearm Dispossession Laws on Household Firearm Ownership



Source: RAND State-Level Household Firearm Ownership Database, 1990–2018.
Notes: The figure plots event-study estimates of the effect of firearm dispossession laws on the state household firearm ownership rate, obtained using the Callaway and Sant'Anna (2021) estimator with not-yet-treated states as the comparison group. The omitted period is the year before enactment. Points are point estimates and vertical bars are 95 percent confidence intervals; standard errors are clustered by state.

Figure 6 displays the event study estimates of the effect of firearm dispossession laws on the household firearm ownership rate. The pre-treatment coefficients are small and statistically indistinguishable from zero, showing no evidence of differential pre-trends and supporting the identifying assumption. Meanwhile, the post-treatment effects are close to zero in the short run and turn only modestly negative around the fifth year after enactment. These estimates suggest firearm dispossession laws do not have a substantial impact on state-level household firearm ownership rates.

Table 5 supports this interpretation of the event study results. The table reports the estimated effect of firearm dispossession laws on the household firearm ownership rate. The

two-way fixed-effects estimate in column (1) implies that enactment reduces household ownership by about two percentage points, significant at the ten percent level. The Callaway and Sant'Anna (2021) estimator in column (2), which accounts for the staggered timing of adoption, yields an effect that is essentially zero and statistically insignificant. We treat this as our preferred estimate: with staggered adoption and heterogeneous treatment effects, the two-way fixed effects specification draws on comparisons between later- and already treated states that can receive negative weights, so the coefficient may reflect this contamination rather than a true reduction in ownership.

Table 5: Effect of Firearm Dispossession Laws on Household Firearm Ownership Rate

	TWFE	CS-DiD
ATT	-0.02*	-0.00
	(0.01)	(0.01)
Observations	702	679

Notes: The dependent variable is the state household firearm ownership rate. Column (1) reports two-way fixed-effects estimates; column (2) reports the Callaway and Sant'Anna (2021) overall ATT, with not-yet-treated states as the comparison group. Standard errors, clustered by state, are in parentheses.
Source: RAND State-Level Household Firearm Ownership Database, 1990–2018.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Taken together, the evidence indicates that dispossession laws tied to DVROs do not measurably shift aggregate firearm ownership at the state level. As previously mentioned, our data do not allow us to measure the effects of these laws on the targeted population of individuals subject to a restraining order. Across the six treated states for which we observe protection order filings, orders are sought at a rate of roughly 210 per 100,000 residents per year (own calculations, over our sample period), and nationally civil protection orders were filed at roughly 380 per 100,000 over 2004–2010.[8] Either way, the covered group is well under one percent of the population—far below the share of households that own firearms—so removing their firearms need not register in a statewide ownership rate.

[8]Own calculations from the NCSC Court Statistics Project, covering 36–46 reporting states annually.

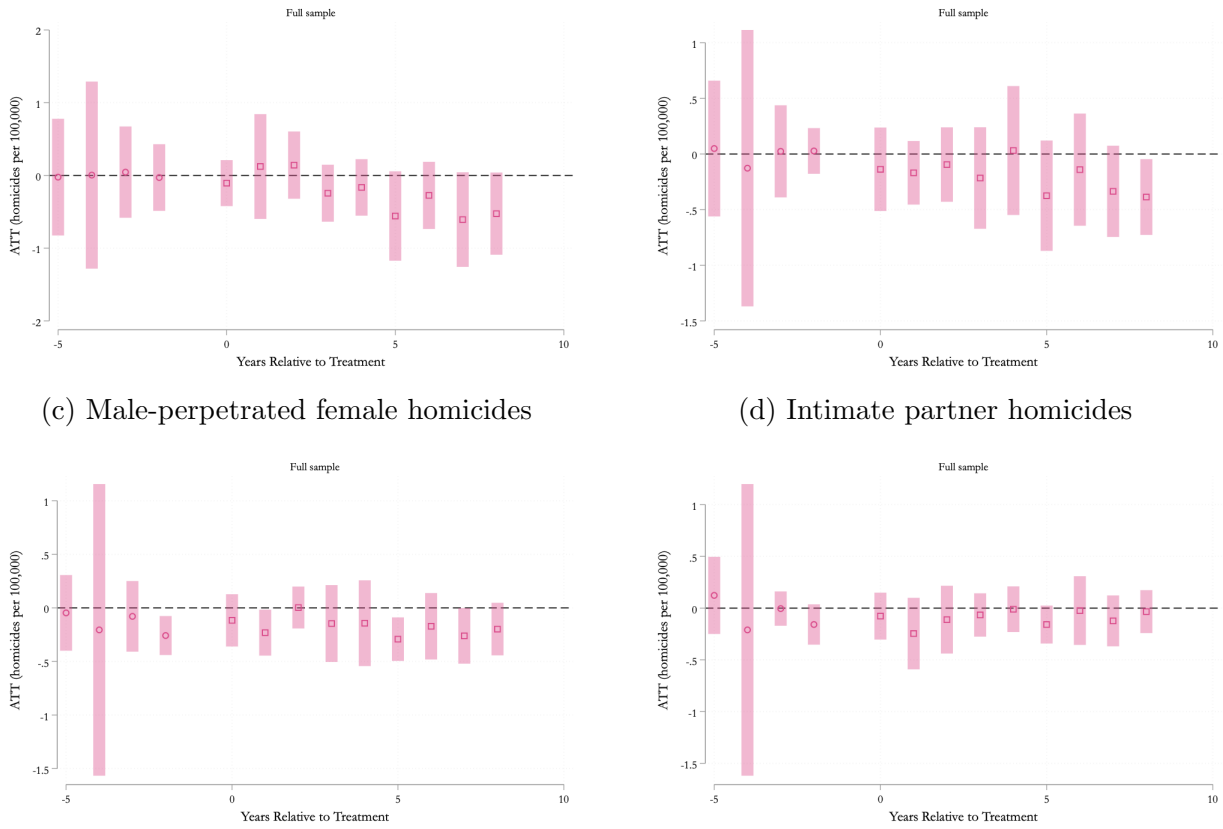
6.2 Effects on homicides

We next examine the effect of dispossession laws on homicides. We consider two victim definitions: homicides committed by an intimate partner, and homicides perpetrated by a male offender against a female victim. For each outcome, we estimate effects separately for homicides committed with a firearm and for homicides committed by any weapon. In every specification, the comparison group is the set of not-yet-treated states, and we report the Callaway and Sant'Anna (2021) overall ATT, controlling for demographic and policing covariates and other contemporaneous laws. Because we cluster at the state level and have fewer than 30 clusters, standard errors are computed using a wild cluster bootstrap.

Figure 7: Event-Study Estimates: Effect of Firearm Dispossession Laws on Homicide

(a) Male-perpetrated female homicides

(b) Intimate partner homicides



(c) Male-perpetrated female homicides

(d) Intimate partner homicides

Source: FBI Supplementary Homicide Reports, 1990–2018.

Notes: Event-study estimates from the Callaway and Sant'Anna (2021) estimator, with not-yet-treated states as the comparison group. The top panel shows homicides per 100,000 population, by any weapon. The bottom panel shows homicides per 100,000 population, by firearms. The omitted period is the year before enactment. Vertical bars are 95 percent confidence intervals; standard errors from a wild cluster bootstrap, clustered by state.

Figure 7 presents the event-study estimates of the effect of firearm dispossession laws on homicides. For every outcome, the pre-treatment coefficients are small and statistically indistinguishable from zero, with no systematic trend in the years leading up to enactment; this supports the identifying assumption that treated and comparison states were on parallel paths absent the law. The post-treatment estimates are close to zero for homicides committed with any weapon in the first years after enactment and grow more negative over the medium run, with the largest reductions appearing roughly five to eight years out. This gradual,

accumulating pattern is consistent with the negative overall ATTs in Table 6 and with a policy whose effect builds as enforcement and compliance take hold rather than appearing immediately.

Table 6: Effect of Firearm Dispossession Laws on Homicides

	Any weapon		Firearms	
	(1)	(2)	(3)	(4)
	Male-Perp. Female Homicides	Intimate Partner Homicides	Male-Perp. Female Homicides	Intimate Partner Homicides
ATT	-0.14	-0.16*	-0.16***	-0.08
	(0.16)	(0.10)	(0.06)	(0.07)
Observations	5713	5713	5713	5713

Notes: Each cell reports the Callaway and Sant'Anna (2021) overall ATT, with not-yet-treated states as the comparison group, controlling for demographic and policing covariates and other contemporaneous laws. Standard errors from a wild cluster bootstrap, clustered by state, in parentheses. Outcomes are per 100,000 population.
Source: FBI Supplementary Homicide Reports, 1990–2018.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6 reports the estimates of the overall ATTs. Columns 1-2 report the effect on homicides by any weapon, and columns 3-4 restrict to firearm homicides. The estimated effects are negative across all outcomes and are economically large. Male-perpetrated female homicides by any weapon fall by 0.14 per 100,000 people (about 18 percent), and intimate partner homicides by 0.16 per 100,000 people (about 24 percent, significant at the ten percent level). Restricting to firearms, male-perpetrated female homicides fall by about 35 percent (significant at the one percent level) and intimate-partner homicides by about 20 percent. Although not every estimate is statistically significant, the uniformly negative sign and comparable magnitude across outcomes, weapons, and subsamples consistently point to sizable reductions in lethal violence.

Substitution between firearms and other weapons. A concern with any law that restricts firearm access is that it may displace lethal violence toward other weapons rather than prevent it. Our estimates bear on this directly. Because the effect on all homicides is the

sum of the effects on firearm and non-firearm homicides, the implied effect on non-firearm homicides is the difference between the any-weapon and firearm estimates in Table 6. For male-perpetrated female homicide, this difference is 0.02 per 100,000 (the -0.14 all-weapon estimate less the -0.16 firearm estimate), and for intimate partner homicide it is -0.08 (-0.16 less -0.08). Estimated directly, neither non-firearm effect is statistically distinguishable from zero.

Enforcement of firearm dispossession laws. The ability and willingness of courts and local law enforcement to enforce firearm dispossession laws may influence their effectiveness. If there is no enforcement mechanism to ensure a DVRO respondent relinquishes and refrains from accessing their firearms, they have little incentive to do so. Ideally, we would replace our treatment indicator in Equation 1 with a measure of enforcement, but the data necessary to do so do not exist. Instead, we exploit the variation in the design of firearm dispossession laws to gauge the importance of enforcement. Specifically, we restrict our sample to states where the law does not permit judicial discretion in determining if a DVRO respondent has to be dispossessed of their firearms. Presumably, the laws are being properly enforced in these states, since judges are legally required to order dispossession.

Figure 8 and Table 7 report the event study and overall ATT estimates, respectively, of the effect of firearm dispossession laws on homicide rates in states where judges *shall* order dispossession for DVRO respondents. The event studies show no evidence of differential pre-trends, consistent with the parallel trends assumption. The post-treatment estimates for male-perpetrated female homicides with any weapon show a gradual decline over time, but the estimates are imprecise. For the other outcomes, the firearm dispossession laws have no apparent effect, though the post-treatment estimates are generally negative.

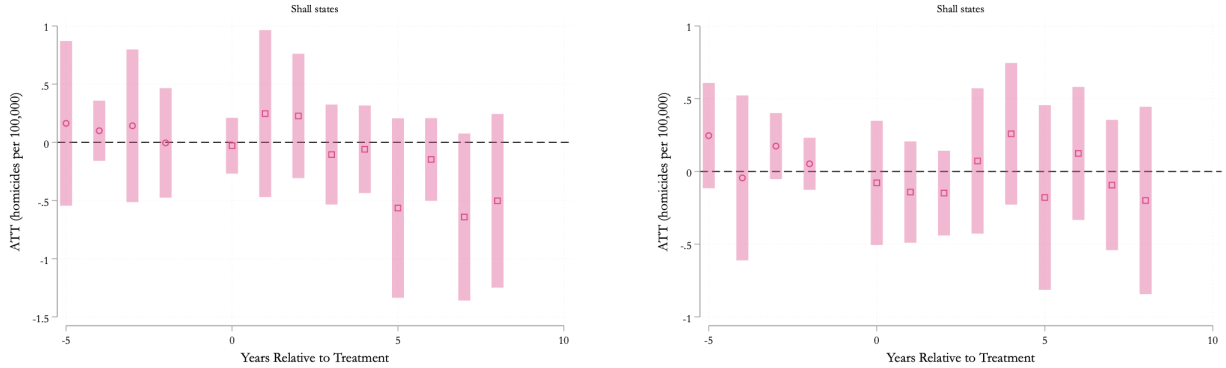
Table 7 tells the same story. All four estimates are negative, and—consistent with a disarmament channel—the firearm-specific point estimates (-0.12 for both victim definitions) are larger in magnitude than the all-weapon estimates (-0.03 and -0.04). None, however, is

statistically distinguishable from zero. With fewer treated states than the pooled sample, the shall estimates are less precise, and their confidence intervals are wide enough to encompass both no effect and economically meaningful reductions.

Figure 8: Event-Study Estimates: Effect of Firearm Dispossession Laws on Homicide, Shall States

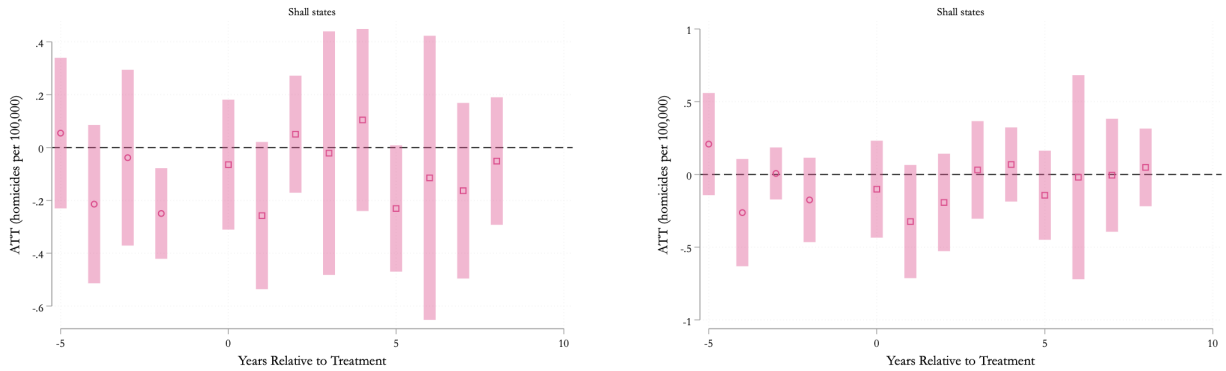
(a) Male-perpetrated female homicides

(b) Intimate partner homicides



(c) Male-perpetrated female homicides

(d) Intimate partner homicides



Source: FBI Supplementary Homicide Reports, 1990–2018.

Notes: Event-study estimates from the Callaway and Sant'Anna (2021) estimator, with not-yet-treated states as the comparison group. The top panel shows homicides per 100,000 population, by any weapon. The bottom panel shows homicides per 100,000 population, by firearms. The omitted period is the year before enactment. Vertical bars are 95 percent confidence intervals; standard errors from a wild cluster bootstrap, clustered by state.

Table 7: Effect of Firearm Dispossession Laws in States with Mandatory Relinquishment

	Any weapon		Firearms	
	(1)	(2)	(3)	(4)
	Male-Perp. Female Homicides	Intimate Partner Homicides	Male-Perp. Female Homicides	Intimate Partner Homicides
ATT	-0.04	-0.03	-0.12	-0.12
	(0.21)	(0.14)	(0.12)	(0.12)
Observations	5423	5423	5423	5423

Notes: Sample restricted to treated states that mandate firearm surrender once a qualifying order is issued ("shall" states). Each cell reports the Callaway and Sant'Anna (2021) overall ATT, with not-yet-treated states as the comparison group, controlling for demographic and policing covariates and other contemporaneous laws. Standard errors from a wild cluster bootstrap, clustered by state, in parentheses. Outcomes are per 100,000 population. *Source*: FBI Supplementary Homicide Reports, 1990–2018. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

6.3 Effects on protection order filings

We next estimate the effect of firearm dispossession laws on civil protection order filings. The outcome is the log of protection order filings. Identification comes from the six states that adopt a dispossession law over the 2003–2018 period during which filings are observed, relative to states that do not adopt.

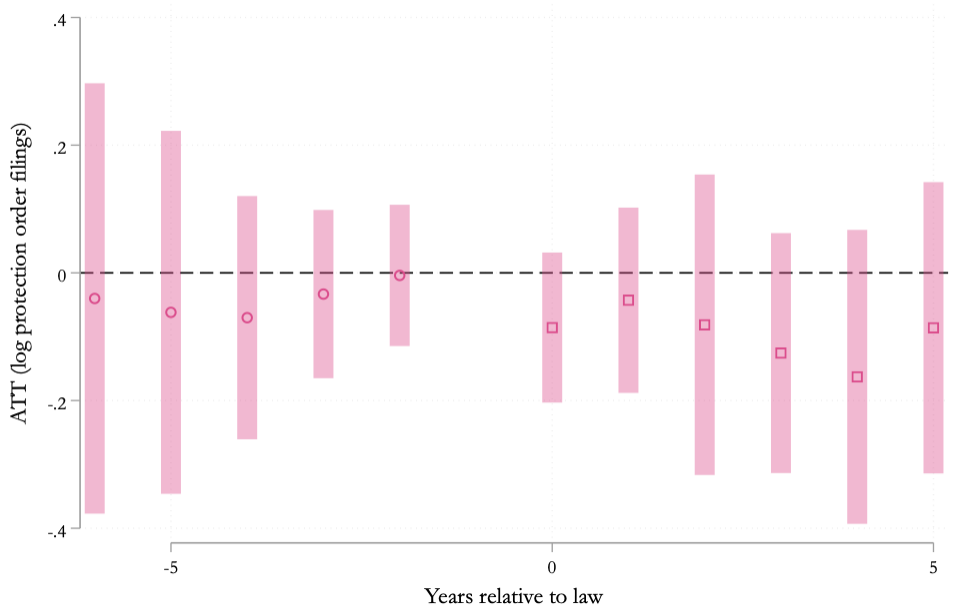


Figure 9: Event-Study Estimates: Effect of Firearm Dispossession Laws on Protection Order Filings

Source: NCSC Court Statistics Project and state court reports, 2003-2018.

Notes: Event-study estimates from the Callaway and Sant'Anna (2021) estimator, with never-treated states as the comparison group. Vertical bars are 95 percent confidence intervals; standard errors from a wild cluster bootstrap, clustered by state.

Table 8: Effect of Firearm Dispossession Laws on Protection Order Filings

	(1) CS-DiD (never-treated)	(2) CS-DiD (not-yet-treated)	(3) TWFE
ATT	-0.067	-0.071	-0.031
	(0.052)	(0.049)	(0.063)
Observations	173	173	189

Notes: The table reports the estimated effect of firearm dispossession laws on civil protection order filings over 2003–2018. All columns use the log of filings as the outcome. All coefficients are interpretable as proportional effects. Columns (1) and (2) report the overall average treatment effect on the treated from the doubly robust estimator of Callaway and Sant'Anna (2021). Column (1) uses never-treated states as the control, while Column (2) uses not-yet-treated states as the control. Standard errors are calculated using a wild cluster bootstrap clustered by state.
Source: National Center for State Courts Court Statistics Project and state court administrative reports, 2003–2018.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 8 reports the results. The Callaway and Sant'Anna (2021) estimate is -0.07 and is not statistically significant. The event-study estimates in Figure 9 show no systematic change in filings in the years before adoption and none after it. Column (2) shows results using the same estimator, but with states not-yet-treated as the control group, and Column (3) shows results using two-way fixed effects. The estimators produce point estimates of different magnitudes and agree that the effect on filings is small and imprecisely estimated.

Identification rests on six treated states, and the confidence intervals do not exclude moderate effects in either direction. We therefore read the filings results as most informative about the mechanism behind the homicide estimates, rather than as a precise estimate of the effect of the laws on protection order activity.

6.4 Discussion of results

Taken together, these form a coherent pattern: dispossession laws reduce lethal intimate partner violence, with the strongest evidence for firearm male-perpetrated female homicide,

while state-level firearm ownership is unchanged and protection order filings are unchanged. The scale of the policy reconciles these findings. Individuals subject to a restraining order constitute well under one percent of the population in a given year, a group too small for its disarmament to register in a state's aggregate ownership rate. These same individuals, however, are substantially overrepresented among the perpetrators of intimate partner homicide.[9] Restricting their access to firearms can therefore reduce lethal violence without altering aggregate ownership.

The protection order results point to the same channel. If dispossession laws reduced homicide primarily by increasing victims' demand for court protection, the number of orders filed would rise after adoption. The estimated effect on filings is instead small and statistically insignificant (Section 6.3), consistent with the laws operating through the disarmament of individuals already subject to an order rather than through a change in the number of people who seek one. Identification here rests on six adopting states, so the protection order filing estimates are imprecise, and we treat them as supportive rather than decisive.

The magnitude of these reductions is consistent with the existing literature. Vigdor and Mercy (2006) report a 7 percent reduction in female intimate partner homicide following the adoption of restraining-order firearm laws, and Zeoli and Webster (2010) report reductions of 19 percent in intimate partner homicide and 25 percent in firearm intimate partner homicide. Our intimate-partner estimates fall within this range, and our estimate for firearm male-perpetrated female homicide lies at its upper end. Because the number of treated clusters is small, our point estimates are imprecise, and we interpret the evidence as a consistent, firearm-specific reduction in lethal violence rather than as a precise effect size.

[9]Roughly six percent of firearm intimate partner homicides involve an offender against whom the victim had sought or held an order (Lyons et al., 2021); this figure is a lower bound, as it counts only orders held against the eventual perpetrator. Prior physical abuse precedes approximately seventy percent of intimate partner femicides (Campbell et al., 2003).

7 Conclusion

This paper asks whether laws restricting firearm access for individuals subject to a domestic violence restraining order reduce lethal intimate partner violence, and through what channel. Using FBI homicide records, a state-year panel of protection order filings, and a staggered difference-in-differences design robust to heterogeneous treatment timing, we estimate the effect of adopting a firearm dispossession law on homicide, on protection order filings, and on firearm ownership.

The estimated effects on homicide are negative across every outcome and are largest, and most precisely estimated, for firearm homicides of women by male offenders, which fall by about 35 percent. Intimate partner homicides fall by roughly a fifth. The reductions are concentrated in firearm homicides and are not accompanied by any increase in homicides committed with other weapons, so the laws appear to prevent lethal violence rather than displace it to other means. These magnitudes are consistent with the existing literature: Vigdor and Mercy (2006) report a 7 percent reduction in female intimate partner homicide, and Zeoli and Webster (2010) report reductions of 19 percent in intimate partner homicide and 25 percent in firearm intimate partner homicide.

Two null results clarify how the laws work. State-level household firearm ownership is unchanged, and so is the number of protection orders filed. Both are consistent with a disarmament channel operating on a narrow, high-risk group: individuals subject to a restraining order are well under one percent of the population, too small a share for their disarmament to move a statewide ownership rate, yet substantially overrepresented among those who commit intimate partner homicide. That filings do not rise after adoption suggests the laws reduce homicide by disarming individuals already subject to an order, not by drawing more victims to seek court protection. Our heterogeneity analysis, however, cannot pin down which statutory design matters more: the discretionary-relinquishment states are too few to be credibly identified, and the mandatory-relinquishment estimates, though uniformly negative and of plausible magnitude, are imprecise.

Several limitations bound what we can conclude. Identification rests on a small number of treated states, so our point estimates are imprecise and we interpret the evidence as a consistent firearm-specific reduction in lethal violence rather than as a precise effect size. Firearm ownership is observed only at the state level, too coarse to detect disarmament concentrated within the targeted group, and the SHR does not separately identify homicides by former dating partners, which likely understates intimate partner homicide. Whether mandatory surrender outperforms judicial discretion remains an open question that a longer panel or richer administrative data on enforcement could help answer. With these caveats, the evidence indicates that restricting firearm access for people subject to a domestic violence restraining order reduces the most lethal form of intimate partner violence, and does so by disarming a small, high-risk population rather than by reshaping gun ownership or the demand for protection orders.

References

Aizer, A. and Bó, P. D. (2009). Love, hate and murder: Commitment devices in violent relationships. *Journal of Public Economics*, 93(3):412–428.

Billings, K. C. and Daniels, M. (2026). Family violence prevention and services act (fvpsa): Background and funding. CRS Report R42838, Congressional Research Service.

Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of econometrics*, 225(2):200–230.

Campbell, J. (2022). Brady handgun violence prevention act.

Campbell, J. C., Glass, N., Sharps, P. W., Laughon, K., and Bloom, T. (2007). Intimate partner homicide: Review and implications of research and policy. *Trauma, Violence & Abuse*, 8(3):246–269.

Campbell, J. C., Webster, D., Koziol-McLain, J., Block, C., Campbell, D., Curry, M. A., Gary, F., Glass, N., McFarlane, J., Sachs, C., Sharps, P., Ulrich, Y., Wilt, S. A., Manganello, J., Xu, X., Schollenberger, J., Frye, V., and Laughon, K. (2003). Risk factors for femicide in abusive relationships: results from a multisite case control study. *American Journal of Public Health*, 93(7):1089–1097.

Capaldi, D. M., Knoble, N. B., Shortt, J. W., and Kim, H. K. (2012). A systematic review of risk factors for intimate partner violence. *Partner Abuse*, 3(2):231–280.

Chin, Y.-M. and Cunningham, S. (2019). Revisiting the effect of warrantless domestic violence arrest laws on intimate partner homicides. *Journal of Public Economics*, 179:104072.

Corporation, R. (2025). State firearm law navigator. Accessed February 28,2026.

Criminal Justice Information Services Division (2026). Domestic relationships and violent crimes, 2020-2024. Technical report, Federal Bureau of Investigation, U.S. Department of Justice.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277. Themed Issue: Treatment Effect 1.

Harrell, E., Truman, J., Fowler, K., Holland, K., Simon, T., and Sumner, S. (2024). *Trends and Patterns in Firearm Violence, 1993-2023*. US Department of Justice, Office of Justice Programs, Bureau of Justice.

Holt, V. L., Kernic, M. A., Lumley, T., Wolf, M. E., and Rivara, F. P. (2002). Civil protection orders and risk of subsequent police-reported violence. *JAMA*, 288(5):589–594.

Holt, V. L., Kernic, M. A., Wolf, M. E., and Rivara, F. P. (2003). Do protection orders affect the likelihood of future partner violence and injury? *American Journal of Preventive Medicine*, 24(1):16–21.

Kaplan, J. (2019). Apparent per capita alcohol consumption: National, state, and regional trends 1977-2016.

Kaplan, J. (2024). *Uniform Crime Reporting (UCR) Program Data: An Opinionated Guide to FBI Data*.

Koppa, V. (2024). Can information save lives? Effect of a victim-focused police intervention on intimate partner homicides. *Journal of Economic Behavior & Organization*, 217:756–782.

Lynch, K. R., Boots, D. P., Jackson, D. B., and Renzetti, C. M. (2022). Firearm-related abuse and protective order requests among intimate partner violence victims. *Journal of interpersonal violence*, 37(15-16):NP12973–NP12997.

Lyons, V. H., Adhia, A., Moe, C., Kernic, M. A., Rowhani-Rahbar, A., and Rivara, F. P. (2021). Firearms and protective orders in intimate partner homicides. *Journal of family violence*, 36(5):587–596.

Miller, A. R. and Segal, C. (2019). Do female officers improve law enforcement quality?: Effects on crime reporting and domestic violence. *The Review of Economic Studies*, 86(5 (310)):pp. 2220–2247.

Pei, Z., Pischke, J.-S., and Schwandt, H. (2019). Poorly measured confounders are more useful on the left than on the right. *Journal of Business & Economic Statistics*, 37(2):205–216.

Sant'Anna, P. H. and Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of econometrics*, 219(1):101–122.

Sharps, P., Campbell, J. C., Campbell, D., Gary, F., and Webster, D. (2003). Risky mix: Drinking, drug use, and homicide. *National Institute of Justice Journal*, (250):8–13.

Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.

ThoughtCo. (2025). Timeline of gun control in the united states.

Vigdor, E. R. and Mercy, J. A. (2006). Do laws restricting access to firearms by domestic violence offenders prevent intimate partner homicide? *Evaluation review*, 30(3):313–346.

Wing, C., Freedman, S. M., and Hollingsworth, A. (2024). Stacked difference-in-differences. Technical report, National Bureau of Economic Research.

Zeoli, A. M., Frattaroli, S., Roskam, K., and Herrera, A. K. (2019). Removing firearms from those prohibited from possession by domestic violence restraining orders: A survey and analysis of state laws. *Trauma, Violence, & Abuse*, 20(1):114–125.

Zeoli, A. M. and Webster, D. W. (2010). Effects of domestic violence policies, alcohol taxes and police staffing levels on intimate partner homicide in large us cities. *Injury Prevention: Journal of the International Society for Child and Adolescent Injury Prevention*, 16:1475–5785.

A Appendix

A.1 Protection Order Terminology

Table 9

Term	Alternative Name	Explanation
Intimate Partner Violence (IPV)	Domestic Violence (DV)	Physical, sexual, emotional, economic, psychological, or technological actions or threats intended to exert control or power over an intimate partner. Intimate partners include current and former spouses and dating partners.
Domestic Violence Restraining Order (DVRO)	Civil Protection Order (CPO)	A civil court order intended to protect an individual from abuse and/or harassment from someone with whom they have a domestic relationship. Domestic relationships generally include intimate partners, family members, and people who share a child in common. Provisions in DVROs can include ordering the ceasing of abuse and contact, support or restitution, relinquishment of firearms, treatment, exclusive use of property, and temporary child custody.
Temporary Restraining/Protection Order (TRO/TPO)	Ex Parte Order	A DVRO issued for a limited amount of time that does not require the respondent be notified or given a chance to respond to the allegations.
Final Order	Permanent Order	A DVRO issued after the respondent is given notice of the petition for a restraining order and an opportunity to participate in a hearing. These orders are effective for a specified amount of time (e.g., one year).
Extreme Risk Protection Order (ERPO)	Gun Violence Restraining Order (GVRO); Lethal Violence Protection Order	A civil court order that temporarily prohibits firearm possession and purchasing for an individual who poses a serious risk of harming themselves or others.
Criminal Protection Order		A criminal court order intended to protect a victim or witness of a crime. These are typically instituted during criminal proceedings, including those involving domestic violence.

A.2 Stacked Difference-in-Differences

This section reports the results from estimating Equation 1 using the stacked difference-in-differences estimator created by Wing et al. (2024). Unlike Callaway and Sant'Anna (2021), this estimator does not utilize all treatment cohorts. Rather, it imposes the restriction that all treated units be observed for the same number of relative periods before and after treatment. We restricted the sample to units observed for at least five periods before treatment and at least eight periods after treatment (i.e., the units treated between 2002 and 2010).

Figure 10: The Effect of Firearm Dispossession Laws on Homicides

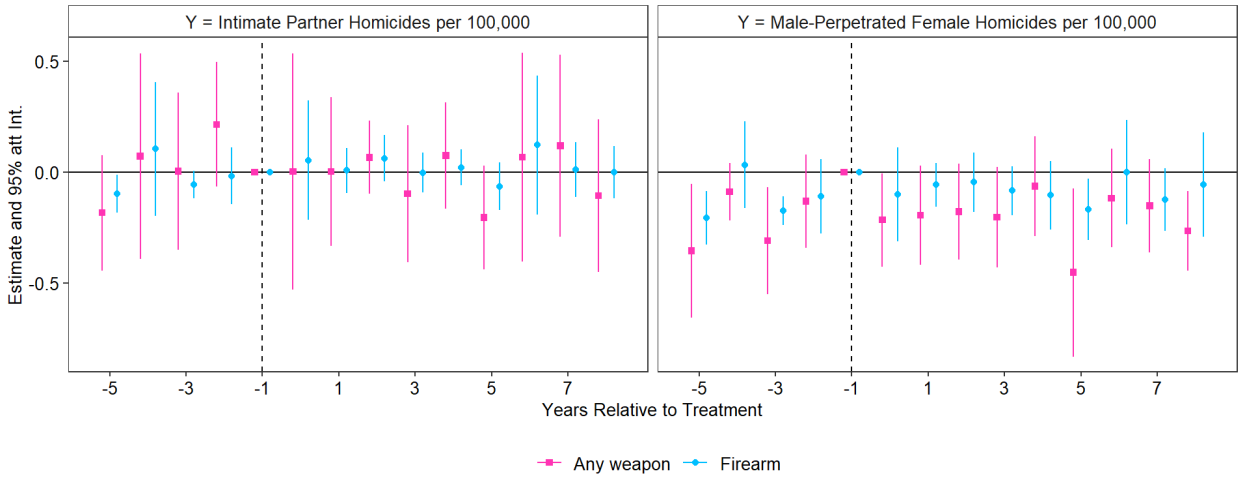
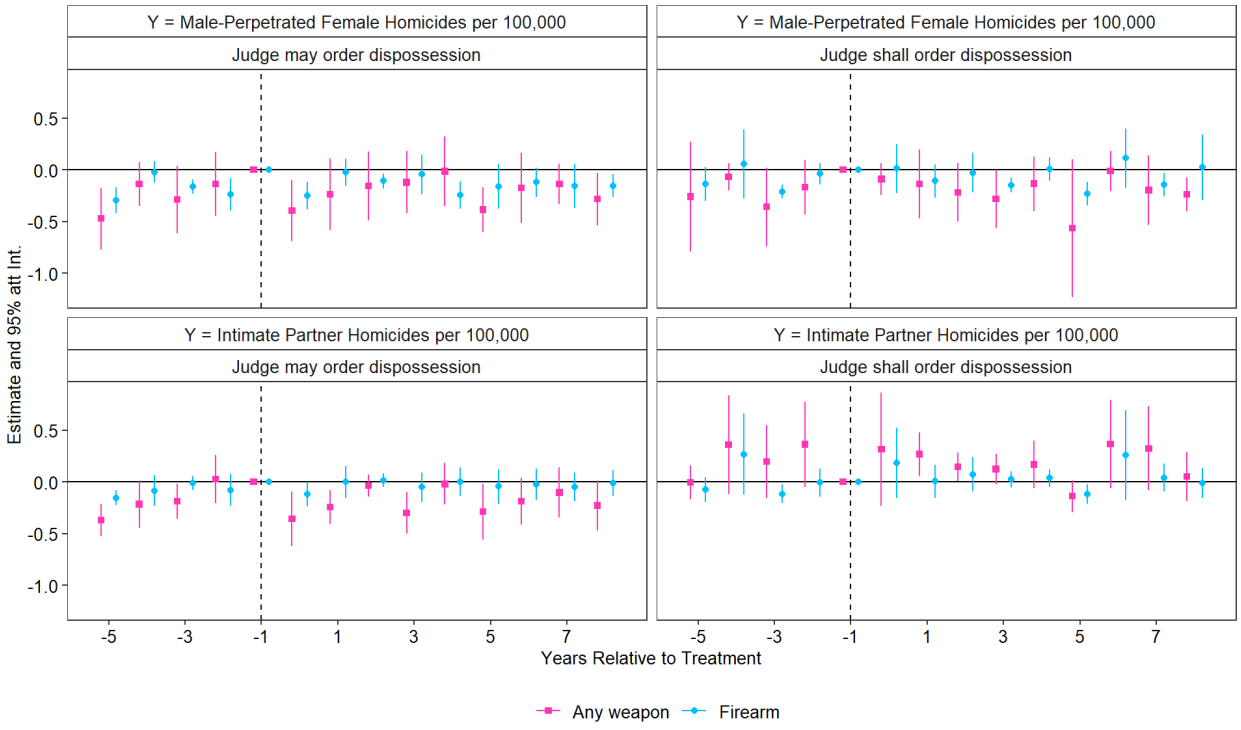


Figure 11: The Effect of Firearm Dispossession Laws on Homicides by Judge's Discretion



Weapon:	Any weapon				Firearms			
Dependent variable:	Male-Perpetrated Female Homicides per 100,000		Intimate Partner Homicides per 100,000		Male-Perpetrated Female Homicides per 100,000		Intimate Partner Homicides per 100,000	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Any dispossession law								
ATT	-0.205**	-0.198**	-0.009	0.000	-0.083	-0.074*	0.023	0.031
	(0.088)	(0.088)	(0.154)	(0.141)	(0.054)	(0.044)	(0.055)	(0.045)
Judges may order dispossession								
ATT	-0.213	-0.192	-0.196**	-0.163*	-0.142***	-0.113*	-0.032	0.000
	(0.130)	(0.126)	(0.086)	(0.088)	(0.052)	(0.059)	(0.048)	(0.050)
Judges shall order dispossession								
ATT	-0.210*	-0.222*	0.181*	0.172*	-0.056	-0.062	0.055	0.047
	(0.127)	(0.133)	(0.108)	(0.095)	(0.059)	(0.049)	(0.066)	(0.060)
Demographic and policing controls	No	Yes	No	Yes	No	Yes	No	Yes
Policy controls	No	Yes	No	Yes	No	Yes	No	Yes
Unemployment rate	No	Yes	No	Yes	No	Yes	No	Yes
County fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,886	11,886	11,886	11,886	11,886	11,886	11,886	11,886

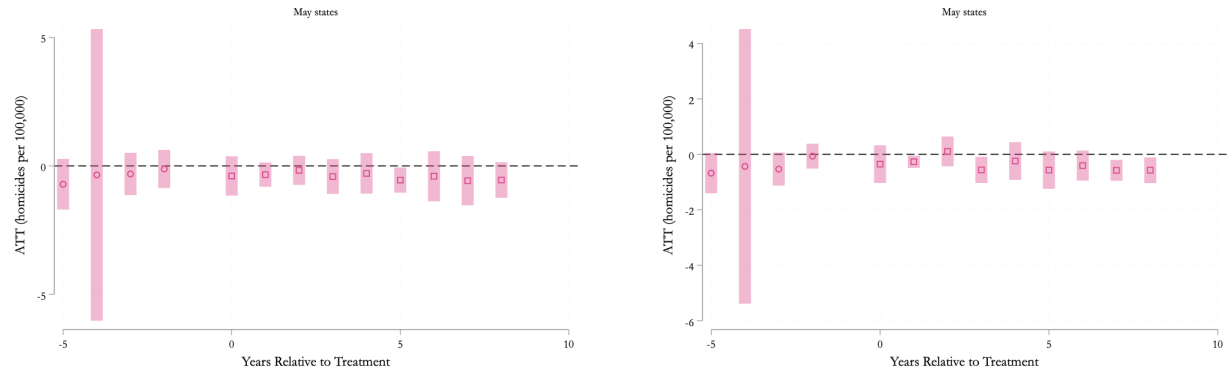
***p<0.01, **p<0.05, *p<0.1

A.3 Event-Study Graphs Using CS-DiD: May States

Figure 12: Event-Study Estimates: Effect of Firearm Dispossession Laws on Homicide

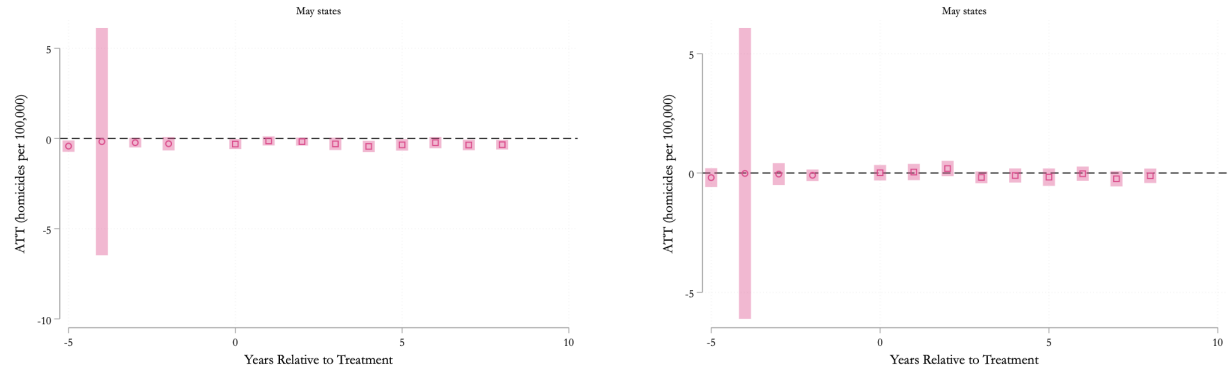
(a) Male-perpetrated female homicides

(b) Intimate partner homicides



(c) Male-perpetrated female homicides

(d) Intimate partner homicides



Source: FBI Supplementary Homicide Reports, 1990–2018.
Notes: Event-study estimates from the Callaway and Sant'Anna (2021) estimator, with not-yet-treated states as the comparison group. The top panel shows homicides per 100,000 population, by any weapon. The bottom panel shows homicides per 100,000 population, by firearms. The omitted period is the year before enactment. Vertical bars are 95 percent confidence intervals; standard errors from a wild cluster bootstrap, clustered by state.

A.4 The Callaway and Sant'Anna Estimator

We estimate the effect of firearm dispossession laws using the doubly robust estimator developed by Callaway and Sant'Anna (2021) (hereafter CS). When treatment timing is staggered, the two-way fixed effects (TWFE) estimator recovers a weighted average of treatment effects in which some of the weights can be negative, because it implicitly uses already-treated counties as controls for counties treated later. If the effect of a dispossession law varies across

adoption cohorts or changes over time, these "forbidden comparisons" can bias the TWFE estimate, in extreme cases producing an estimated effect with the opposite sign of every underlying group's true effect (Goodman-Bacon, 2021). The CS estimator avoids this problem by construction: it never uses already-treated counties as controls, instead comparing each treated cohort only to counties that are not yet treated or never treated, and it estimates a separate treatment effect for each cohort in each period before aggregating with transparent, non-negative weights.

Group-time average treatment effects. The fundamental building block of the CS estimator is the *group-time average treatment effect*. We index counties by their *group*, G_c, defined as the first year in which county c's state has a firearm dispossession law in effect. Counties whose states never enact a dispossession law during the sample period serve as the never-treated comparison group, for which we write $C_c = 1$. For a group g and a period t, the group-time average treatment effect is

$$ATT(g,t) = \mathbb{E}[\, Y_{ct}(g) - Y_{ct}(0) \mid G_c = g \,], \tag{3}$$

where $Y_{ct}(g)$ is the potential homicide outcome if county c is first treated in year g and $Y_{ct}(0)$ is the potential outcome if c is never treated. $ATT(g,t)$ is the average effect of having a dispossession law at time t for the counties whose states first adopted one in year g. Unlike the single coefficient produced by a TWFE regression, CS leaves these cohort-by-period effects disaggregated, so that heterogeneity across cohorts and over time is preserved rather than averaged away with potentially negative weights.

Identification. The CS estimator identifies $ATT(g,t)$ under three assumptions analogous to those discussed in Section 5.1: (i) *no anticipation*, that counties experience no treatment effect in the years before their state's law takes effect; (ii) *conditional parallel trends*, that absent treatment the treated and never-treated counties would have followed the same trend

in the outcome conditional on the covariates X_{cst}; and (iii) *overlap*, that for every covariate value a positive fraction of counties remain in the never-treated comparison group. Under these assumptions, and using the never-treated counties as the comparison group, $ATT(g,t)$ is identified for $t \geq g$ by the doubly robust estimand of Sant'Anna and Zhao (2020),

$$ATT(g,t) = \mathbb{E}\left[\left(\frac{\mathbb{1}\{G_c = g\}}{\mathbb{E}[\mathbb{1}\{G_c = g\}]} - \frac{\dfrac{p_g(X_{cs})\, C_c}{1 - p_g(X_{cs})}}{\mathbb{E}\left[\dfrac{p_g(X_{cs})\, C_c}{1 - p_g(X_{cs})}\right]} \right) \Big(Y_{ct} - Y_{c,g-1} - m_{g,t}(X_{cs}) \Big) \right], \quad (4)$$

where $p_g(X_{cs})$ is the propensity score for being in group g (versus never-treated) given the covariates, and $m_{g,t}(X_{cs}) = \mathbb{E}[\, Y_{ct} - Y_{c,g-1} \mid X_{cs}, C_c = 1\,]$ is the outcome regression for the change in the outcome among never-treated counties. The estimand in Equation 4 is "doubly robust" in that it consistently estimates $ATT(g,t)$ if *either* the propensity score model *or* the outcome regression is correctly specified, but not necessarily both. Each $ATT(g,t)$ is estimated by replacing the population expectations in Equation 4 with their sample analogues.

Aggregation. Equation 4 yields a separate effect for every group g and period t, which is too disaggregated to summarize. Following Callaway and Sant'Anna (2021), we aggregate these estimates into interpretable summary parameters using weights that are always non-negative. To assess the dynamics of the policy, we aggregate by length of exposure $e = t - g$, the number of years since a county's state enacted its law:

$$\theta_{es}(e) = \sum_{g} \mathbb{1}\{g + e \leq T\}\, ATT(g, g+e)\, P(G_c = g \mid G_c + e \leq T). \quad (5)$$

The parameter $\theta_{es}(e)$ is the average effect of a dispossession law e years after enactment, averaged across all cohorts observed at that exposure length, and is the CS analogue of the event-study coefficients β_l in Equation 2. We plot these event-study coefficients, together with simultaneous confidence bands, to display the dynamics of the policy and to assess the pre-treatment parallel-trends assumption.

To summarize the policy's overall effect in a single number, we collapse the group-time effects into a weighted average of all $ATT(g,t)$ in post-treatment periods ($t \geq g$), where each effect is weighted by the relative size of its cohort:

$$\theta_{simple} = \sum_{g} \sum_{t \geq g} ATT(g,t)\, w_{g,t}, \qquad w_{g,t} = \frac{P(G_c = g)}{\sum_{g'} \sum_{t' \geq g'} P(G_c = g')}, \tag{6}$$

so that the weights are non-negative and sum to one. The parameter θ_{simple} is the average effect of having a firearm dispossession law across all treated cohorts and post-treatment years, with larger cohorts and cohorts observed for more post-treatment periods receiving proportionally more weight. This is the simple weighted-average aggregation we report in our results table. We construct the event-study confidence bands and the standard error for θ_{simple} using the multiplier bootstrap procedure of Callaway and Sant'Anna (2021), clustering at the state level.